Ocean & Coastal Management 149 (2017) 210–216

Contents lists available at ScienceDirect

Ocean & Coastal Management

journal homepage: www.elsevier.com/locate/ocecoaman

A game theory approach to fishers' strategic behavior vis-à-vis hotel-based water sports operators – The case of the Balaclava Marine Park project in Mauritius

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ARTICLE INFO

Article history:
Received 23 June 2016
Received in revised form
11 April 2017
Accepted 20 June 2017
Available online 22 July 2017

Keywords:
Fisheries
Marine park
Marine resources
Management of resources
Responsibility
Strategic behaviors
Game theory
Recreation
Mauritius
Island states

ABSTRACT

Using a game theory framework, this paper models the behavior of two stakeholder groups who are concerned with the implementation and enforcement of a marine park at Balaclava, situated in the North-West of the island of Mauritius, located in the Indian Ocean. These two stakeholders are artisanal fishers and hotel-based pleasure craft/water-sports operators who often have to compete spatially within the lagoon to carry out their activity. Nash equilibrium results show that fishers and pleasure craft operators may reach a sub-optimal outcome, leading to unsustainability of marine resources. Empirical verification of strategic motives, attitudes and perceptions, using a questionnaire-based survey, actually revealed that there is little convincing evidence that both parties would be able to collaborate and co-manage the park successfully despite the fact that they agree that a marine park is vital for protecting marine resources. While 81.3 per cent of the fishers and 90.1 per cent of the hotel water-based sports operators were in favor of a marine park, only 28 per cent of the hotel sports operators would be willing to work in collaboration with fishers and 46 per cent of the fishers would be willing to work with their counterparts. One of the major policy recommendations that emanates from this study is that the authorities should come up with an effective action plan to foster more dialogues among all concerned stakeholders and emphasize the virtues of co-management for the sustainability of the park.

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1. Background

Poverty alleviation and environmental protection are often viewed as conflicting objectives in developing countries (Kunzmann, 2004; Rozelle et al., 1997; Scherr, 2000). Particularly in island states and coastal regions, the fisheries sector is often subjected to severe environmental degradation. The latter thus constitutes a direct threat to sustainable livelihoods for the fishing communities. As common property resources, coastal and marine resources are often under the control of multiple stakeholders, who are simply interested in maximizing their own benefits, even if it is to the detriment of the environment. (Sobhee, 2004a, 2006) have shown that beach-based tourism, in the context of Mauritius and small island states, has largely favoured the construction of several hotels and related unsustainable built-up areas along the coastline, thereby impacting severely the coastal and marine environments. On the other hand, there are fishing communities who, due to their large populations, low educational attainment and limited scope for alternative economic activity, tend to overexploit marine resources, eventually depleting fish stocks (Sobhee, 2004b). In addition, there are people who are not registered fishers but who actually contribute to environmental degradation by over-practising fishing as a recreational activity. In fact (Cooke and Cowx, 2004, 2006), reported that, alongside commercial fishing activity, recreational fishing activity is also contributing to the degradation of marine resources. Moreover, there are pleasure craft operators who use high speed boats to carry mainly tourists and local visitors to do para-sailing, wind-surfing, snorkeling, and kiting amongst others. The latter do as well contribute to further environmental degradation by damaging corals and animal habitats.

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http://dx.doi.org/10.1016/j.ocecoaman.2017.06.009
0964-5691/© 2017 Published by Elsevier Ltd.

Clearly, unchecked recreational activity exacerbate environmental damages to coastal and marine resources. In a socioeconomic setting, in which there are several stakeholders involved in sharing marine resources, it is hardly possible to avoid the so-called '*tragedy of the commons*' (Hardin, 1968, 1998). In the absence of any co-operation, it is difficult to reach consensus on how to allocate resources in a fair or equitable manner especially when property rights are ill-defined and multiple stakeholders are involved as indicated by (Coase, 1960). However, to ensure sustainable livelihoods in the long run, the concept of a marine park, as a protected area, could possibly be a solution (see for instance (Suman et al., 1999; Ostrom et al., 2008; Wolfenden et al., 1994) on the benefits of a marine park). It is worth noting that this solution is only viable if and only if all stakeholders actively participate and co-operate in the process of implementation and in the effective management of such a park to optimize collective opportunities and benefits. Co-operation as pointed out by (Ostrom et al., 2008) is primordial and constitutes the social capital in developing countries to circumvent the problems related to the 'tragedy of the commons'. Communication gaps among stakeholders may jeopardize the entire process as emphasized by (Ostrom et al., 2008).

In principle, conflict occurs when there is major disagreement between two or more groups that have a common interest in a given venture. In the present case, conflict between hotel sports operators and fishers is emphasized since the two often have to share the limited marine space as it is really difficult for both parties to carry out their activities simultaneously within the lagoon. Sports operators can provide kite surfing, surfing, snorkeling, fishing and other activities within the lagoon using high speed boats. Fishers on the other hand rely on their artisanal techniques; using mostly, pirogues, fishing baskets and traps to earn a living. Disagreements over the frequency of trips that both parties need to undertake and the intensity of each activity (duration of fishing activity or duration of sports activity during a day) are common. Actually there exists no law that could address any sort of conflict between these two stakeholders when it comes to sharing the marine environment. Everything relies on a mutual understanding and co-operation between the stakeholders.

Marine Protected Areas (MPAs) such as parks can be seen as a possible solution to the "tragedy of the commons" as they are guided by two main principles; they aim at conserving/maintaining biodiversity and intrinsic ecosystem services and contribute towards sustainable fisheries. Dayton et al. (1995) and Bohnsack (1996) make it clear that there are several distinct reasons for establishing MPAs. Among these arguments are the protection and effective management of local marine resources, maintenance of biodiversity and promotion and control of beach-based tourist activity. In fact, MPAs are considered as efficient mechanisms for protecting marine habitats, promoting biodiversity and increasing the stock of species. As such, they help preventing the extinction of species and ensuring sustainable livelihoods of fishers (Dayton et al., 1995; Bohnsack, 1996). According to (Bohnsack, 1996), the effectiveness of marine protected areas depends on the extent to which the objectives are planned. MPAs are also spaces to reduce conflicts, to nurture responsible attitudes among stakeholders and provide a common platform for them to voice out their interest and opinion.

This paper particularly focuses on these aspects and highlight the scope and challenges involved in uniting different stakeholders in the context of a marine protected area. The research is considered from a small island developing state perspective, carried out in the island of Mauritius, located in the Indian Ocean. Local authorities are interested to implement a marine park at Balaclava in the northern part of the island to address the degradation of marine resources, preserve the ecosystem of the area and ensure sustainable exploitation of the resources. The paper also relates to an interesting case study involving two major stakeholders, namely, hotel water-sports operators and the local fishermen. Their strategic behaviors are modeled using game theory, with a little more emphasis on fishers' behavior. A survey is then carried out to provide empirical evidence on the behavior of both stakeholders against the predictions of the theoretical model.

In so doing, this paper extends the literature on ocean and coastal zone management by providing exclusive insights into an island's perspectives. Mauritius is actually a favorite tourist destination[1] for many European and high-income countries. Being of volcanic origin, the island offers wonderful beaches and landscapes, but at the same time, it has very fragile ecosystems. The literature on the public's attitudes towards marine parks remains scanty (see (Suman et al., 1999)) and any country experience may therefore contribute to additional knowledge by capturing specific features (see for instance (Wolfenden et al., 1994; Farrow, 1996; Bazin et al., 2008; Shaffer and Benzaken, 1988)). Such research and findings do help to formulate policies towards sustainable management of marine and coastal resources.

The increasing number of tourists in Mauritius, which exceeded one million in 2016, has led to the construction of several hotels along the coastline -often viewed as unsustainable built-up areas (See (Sobhee, 2004b)). The island has a population of 1.2 million and is categorised as an upper-middle income country with a current per capita income of US $ 9400 in 2016. While tourism is a major income generating recreational activity, it contributes towards employment creation and foreign exchange reserves but has severe repercussions on the physical environment. Damages to corals, removal of mangroves, sand-dredging and over-fishing have largely exacerbated the problem of environmental degradation. The lack of enforcement of environmental laws and regulations has speeded up such degradation. It is within this context that the government has proclaimed Balaclava (a highly touristic area) as a marine protected area (MPA) in the region of Pointe aux Piments in 2007. Since then, little has been done to implement the marine park or enforce it. In recent years, accentuated marine and coastal degradation, and pressure from Non-Governmental Organisations (NGOs), have revived interest in this MPA.

One of the main objectives set for this paper is to analyse the complexity and challenges of co-management of the park that would determine sustainable resource exploitation and sustainable livelihoods in the end. The willingness to work with each other group and the degree of collaboration would clearly provide insights into the success of the participatory process of managing the park once the latter would be established. In this respect, the behavior of two concerned major stakeholders, the registered fishers and the hotel water-sports operators is analysed. The authors words attempt to capture their perceptions and attitudes towards the marine park to determine the robustness of their eventual participation in this process. The rest of the paper comprises a conceptual framework based on theory of games linked to ill-defined property rights that circumscribe the park, followed by a discussion on the location of the protected area and its characteristics. Empirical evidence is also provided based on a survey to test the veracity of the model. The last section provides policy recommendations and concludes. In the present context, the application of game theory is fully justified given that previous studies have been mostly confined to addressing conflicts through survey methods directly and identifying gaps in laws and policies in place. Studies on island economies focus mostly on marine environment

[1] Tourism in Mauritius is essentially beach-based. Among the tourists, most of them are from European countries with France as the leading market.

Please cite this article in press as: Sobhee, S.K., et al., A game theory approach to fishers' strategic behavior vis-à-vis hotel-based water sports operators – The case of the Balaclava Marine Park project in Mauritius, Ocean & Coastal Management (2017), http://dx.doi.org/10.1016/j.ocecoaman.2017.06.009

or coastal environment degradation caused either by over-exploitation of the lagoons and tourism impacts. Moreover, little is said on the strategic behavior of stakeholders especially in the context of establishing the marine park. The analysis is also supported by empirical evidence thereby contributing to further enhance the literature on ocean and coastal management, particularly, for island states. One major advantage of the model is that it allows for various scenarios and strategic motives under varying circumstances.

2. Methodological framework

The authors apply game theory to model the strategic behavior that characterizes the two main stakeholders (henceforth treated as agents) in this game. The rationale for using this model is particularly to capture the strategic motives and moves of agents who compete to access and use the marine resources. The framework also allows for the possibility of co-operation and maximization of returns through some form of collusive agreement or understanding between the two players. However, this agreement does not preclude any form of cheating (that is, violation of any form of the agreement) in order to maximize one's own pay-offs. Game theory allows for different possibilities representing each a specific strategic motive with a certain corresponding outcome.[2] For the sake of convenience, it is assumed that all the fishers can be brought together to form one group while a team of hotel-based recreational sports operators can be formed out of the different hotels in that region. Assuming their moves to be finite and information flow to be imperfect, it is not always possible to keep track of what kind of strategy is being adopted by a rival player.[3] The main objective of the model is to show that, in the absence of co-operation and mutual understanding between the two agents, marine resources would be unsustainably exploited leading to unstable returns for both players. Alternatively, in the event, they stick to their pact and the rules set in the game, the opposite would happen and livelihoods would be sustainable for both parties. The authors propose to analyse two situations; behavior prior to and after the settlement of the marine park. The defined strategies in this game are confined to either co-operate or not co-operate. However, there is no binding agreement between the players to force them to co-operate or default as such. Any agreement to co-operate is simply through a mutual understanding. It is also taken that the behavior of one agent from each group is a representation of the behavior of the all other agents of that particular group. In other words, players in either group have homogeneous behavior. To facilitate the discussion, the authors focus on one specific agent, which happens to be the fisher or the fishers' group.

To understand the fishers' motives, the following framework is put forward:

The profit function (π) of the fishers' group is modeled as:

$$\pi = (p - c)H$$

where p is the average price of fish caught, c is the average cost incurred per unit and H is the harvest rate. This rate can be further written as:

$$H = aE^{\frac{\theta}{R}}, \text{where } E = T \times N$$

E is the fishing effort which is a product of hours spent at sea (T) and number of fishing spots (N). θ is an impact elasticity and R (>0) is a parameter introduced to capture any restriction that emanates from the agreement with the hotel group. Note that when R = 1 this will imply that no restriction exists and whenever R exceeds 1, this will mean that restrictions are imposed (higher R means more restrictions) and this situation tends to reduce the effectiveness of the fishing effort and the return of the fisher as captured by the following partial derivative:

$$\frac{\partial H}{\partial R} = \frac{-\theta}{R^2} H \ln E < 0.$$

2.1. Prior to the creation of the marine protected area

Given this situation, what follows is a reaction of the fisher to the different possible strategies that he may consider to maximize returns irrespective of its pact with the hotel group.

Suppose S_f is denoted as a strategic move of the fisher, defined here precisely as Co-operate and S_h as the strategic move of recreational sports operator to co-operate as well. As such, the complement of S denotes the Not Co-operate strategy, indicated by S'. Each agent aims at maximizing utility from an adopted strategy based on the expected return or pay-off emanating from that strategy. This can be explained by considering the following pay-off matrix for both parties adopting alternative strategies:

$$U_f\left(S'_f, S_h\right) > U_f\left(S_f, S_j\right) > U_f\left(S'_f, S'_h\right) > U_f\left(S_f, S'_h\right)$$

In other words, the highest payoff for the fisher agent is derived when he does not co-operate, while his rival does co-operate. Below is Fig. 1 which summarises all the possible outcomes.

Now, if both agents 'Co-operate' and come to an agreement with respect to exploitation of the marine resources, they will be able to raise their pay-offs as compared to a situation shown in the South East quadrant where they each adopt a 'Not Co-operate' strategy. However, there is an incentive for each player to diverge as he may be better off to choose the alternative strategy, thinking that his counterpart will be unaware. This is typically referred to as the Prisoner's Dilemma. The hotel tour operator has the incentive to choose a 'Not Co-operate' strategy as indicated in the South Western quadrant where his pay-off is the maximum (18). Given this move, the Fisher will choose a Not Co-operate strategy as well with a pay-off of 3 rather than adopting the Co-operate strategy with a pay-off of 2. Similarly, if the fisher

		Fishers	
		Co-operate	*Not Co-operate*
Hotels	*Co-operate*	15 / 5	9 / 7
	Not Co-operate	18 / 2	12 / 3

Fig. 1. Pay-off matrix before creation of the marine protected area.

[2] For a formal review of different strategic motives from a theoretical perspectives, refer to (Gibbons, 1992) (Osborne, 2004), and (Wydick, 2008).

[3] It is also the case where a player may learn what the other player is actually doing after a time lag.

chooses a strategy as a first mover, the sports operator will choose 'Not Co-operate' where he obtains a pay-off of 7 and given this strategy, it will be more profitable for hotel tour operators to choose a strategy of Not Co-operate. Eventually, both parties end up not co-operating and this would result in the lowest outcome in the South Eastern quadrant. This is actually the Nash equilibrium outcome where there is no further incentive to cheat. As both players default, a sub-optimal outcome is obtained, that is, unsustainable resource exploitation.

2.2. The case of uncertainty and mixed strategies

The Nash equilibrium outcome illustrated and discussed in Fig. 1 is also consistent with risks minimization criteria usually applied in game theory exercises characterized by uncertainty. For instance, if the criterion of Max-min is applied to the context of the fisher agent, it would be found that the outcome corresponds to a pay-off of 3 found in the South Eastern quadrant. This holds true for the hotel sports operator whose pay-off turns out to be 12 in the same quadrant. Moreover, if the Minimax Regrets criterion is used, the resulting equilibrium would be Nash-consistent. In the event a weighted-average method is chosen such as that of the Hurwicz's criterion the Nash equilibrium remains the same as explained below:

Under the Hurwicz's approach, where P is the probability of being optimistic, the pay-offs can be expressed as:

$$(1-P)U_f\left(S'_f, S_h\right) + PU_f\left(S'_f, S_j\right) > (1-P)U_f\left(S_f, S'_h\right) + PU_f\left(S_f, S'_h\right)$$

Suppose a probability of ¼ is adopted for being pessimistic (automatically, it follows that ¾ would be the probability of being optimistic). Hence, from the above table, the outcome for the 'Co-operate' strategy would be $5(1/4) + 2(3/4) = 11/4 = 2.75$ while for the 'Not Co-operate' strategy the outcome would be $7(1/4) + 3(3/4) = 16/4 = 4$. The same could be shown for the hotel group, thus indicating that Nash equilibrium is robustly defined even under uncertainty conditions.

All the above equilibrium outcomes and discussions relate to a case where there is nothing like an MPA, but just mere strategies that could be adopted by the two players as stakeholders. In the next section, the idea of an MPA is considered with resulting alterations in adopted strategies and outcomes.

2.3. Creation of a marine protected area

It can now be assumed that the two agents realize that one way they could increase their pay-offs in the long run irrespective of their adopted strategy would be through the creation of a marine protected area, where there is strictly more binding control over the resources in terms of recreational activity and fishing. In principle, commercial and recreational uses are allowed in the park but these would be monitored closely so as not to compromise conservation objectives (Boersma and Parrish, 1999). With such arrangement by the government, the discounted pay-offs in current monetary worth are assumed to double relative to the original matrix.[4] Such a matrix is illustrated below. Here, without loss of generality, the authors abstract from performing a complete discounting exercise and assume that the pay-off for each party under each strategy is denoted by:

[4] The increase in pay-offs may be justified on the ground that the park would bring about an increase in bio-diversity, proliferation of corals, increase in fish stocks within the park and gradually in its vicinity.

		Fishers	
		Co-operate	*Not Co-operate*
Co-operate	**Hotels**	30 / 10	18 / 14
Not Co-operate		36 / 4	24 / 6

Fig. 2. Pay-off matrix after creation of marine protected area with No enforcement.

$$PV(U_i) = \sum_0^n \left(E\frac{U_i}{(1+\partial)^t}\right)$$

$PV(U_i)$ = Present Value of the pay-off of player i.

$\left(E\frac{U_i}{(1+\partial)^t}\right)$ = Discounted expected pay-off of agent i over a given period of time enough to ensure the regeneration of marine resources (fisheries, corals and sea-weeds). For the sake of convenience, the discounted rate is taken to be identical for each agent and assumed to be equal to ∂.

Two cases are considered here; one in which there is very poor or no enforcement by the authorities once the park is created, and another situation in which there are strict rules adopted along with a set penalties in case of default by either player. The authors further assume that, with the creation of the marine park, the two protagonists would stick to a rule of co-operation. But such a tactic is adopted for certain period of time as a co-management strategy and in reciprocity to allow marine resources to replenish in the protected area - albeit not permanently. The two agents have realized that to increase their long term benefits they would need to stick to their commitment. However, once the resources have been allowed to replenish and all concerned parties have stuck to the rule of co-operation, they would adopt new strategies to improve their current benefits. Such behavior would give rise to an intermediate Nash equilibrium which would be short-lived. Hence, once the resources would be replenished, the two players would start over again playing strategic games as before to their advantage. Fig. 2 reports the outcomes of this situation.

In the long run, with little or no enforcement, free riding would again take place followed by the Prisoner's Dilemma driving back the Nash equilibrium to the South Eastern quadrant once again (as shown in Fig. 2). The difference compared to the short run (that is, in the absence of the MPA, Fig. 1)) is that the pay-offs are higher. In fact, this situation is unsustainable as the returns may decrease substantially due to over-exploitation of the lagoon through free-riding.

To remedy the situation, and to ensure that the protected resources are sustainably exploited, it would be better if the government enforces the law by introducing severe penalties on those stakeholders who do not respect their agreement with local authorities. Severe penalties may greatly impact the pay-offs as illustrated in the following matrix Fig. 3:

In Fig. 3 it is shown that whenever agents choose a 'Not Co-operate' strategy, their pay-offs tend to fall because these are due to potential penalties that may be imposed in case of default. This could be in the form of fines or temporary expulsion from the park.

		Fishers	
		Co-operate	*Not Co-operate*
Co-operate	**Hotels**	30 / 10	18 / 4
Not Co-operate		16 / 4	16 / 2

Fig. 3. Pay-off matrix after creation of marine protected area with enforcement and penalties.

Hence, there would be no incentive to cheat and the Nash equilibrium outcome would shift to the North Western quadrant where each party would be urged to adopt a 'Co-operate' strategy. Needless to say, this outcome would be consistent with sustainable resources management.

3. Field experiment and demographic characteristics of agents

As far back as October 1997, 353 hectares (ha) at Blue Bay (South East of the island) and 485 ha at Balaclava (North West of the island) constituting the marine environment (sea-surface area) were proclaimed as National Parks by the Government of Mauritius (2010). The main aim was to adopt a long-term sustainable marine environment policy aiming at the protection and conservation of marine ecosystems for sustainable use.[5] In spite of the fact that Balaclava has been proclaimed as an MPA as far back as 1997, there have been very little efforts done by the government to encourage its enforcement as is the case with Blue Bay MPA. The Balaclava MPA (BMPA) would help to protect marine resources for renewing marine species, avoiding reef degradation and enhancing the habitats. A variety of coastal and marine biotopes exists in the marine parks as well as a diversity of species, some of which are endemic, endangered and thus protected to avert the loss of tropical reef biodiversity. However, 39% of live corals have been affected by bleaching in the lagoons at BMPA which forms part of the marine protected areas within the Indian Ocean Commission Island member states. The marine parks have important socio-cultural, historic, economic aesthetic and ecological values within the Mascarene region, which is among the 18 hotspots in terms of marine biodiversity richness. From the perspectives of this research, an attempt was made to gauge the attitudes and perceptions of two different stakeholder groups of the BMPA in case enforcement takes place and a marine park becomes well implemented. The findings would thus provide guidance to the government and non-governmental organisations for an effective management of the park by private agents. The BMPA is a zone which is of significant economic and ecological importance to the local people and to the island as a whole. It is a site which is an attraction for beach-based tourism. There are at least 15 recognised hotels in the region. They help to generate employment for both the people in the vicinity as well as elsewhere in the country through their direct and indirect (or related) activities. Local fishermen as well as non-hotel based water-sports operators do also benefit from the site in earning a living in addition to the local community interested in outdoor recreation. Hence, degradation of the marine environment circumscribing the MPA would entail serious damages leading to unsustainable tourism and livelihoods for the relevant stakeholders. It is in this context that the study would help to better understand the nature and limitations of any interaction among the different groups which could be potentially involved in the co-management of the park once well established.

The empirical evidence is based on the fieldwork carried out at Pointe aux Piments in 2009, the closest rural village opposite to the BMPA. Fig. 4 below shows the location of the BMPA in the north-west of Mauritius.

The length of the beach opposite the BMPA is approximately 11.7 ha with a sea frontage of 3.6 km (SM, 2012), while the overall length of the coastal line of the entire island of Mauritius is around 100 km. Out of the beach length at Pointe aux Piments village, only 0.2 ha relates to the main beach consisting of 0.3 km. The remaining coastal line in that region is occupied mostly by built-up areas such as hotels and restaurants (15 in number inclusive of those which were being refurbished at the time of the survey) and a fish-landing station located on the main public beach. In fact, there are 16 registered fishermen based in that region and 19 hotel-based water-sports operators. While most of the fishers were found to live in the village of Pointe-aux-Piments, one was a resident outside the

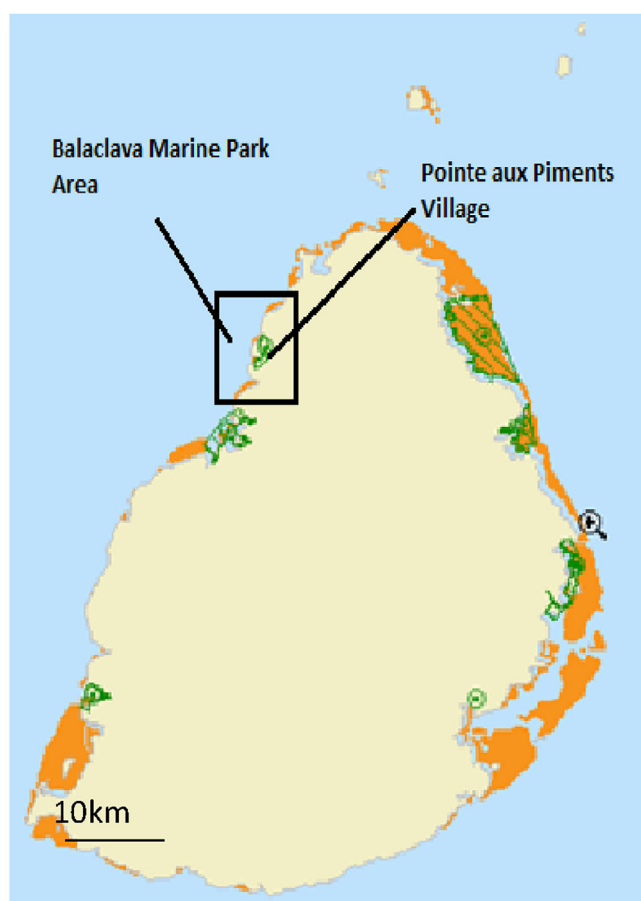

Fig. 4. Balaclava marine protected area in the north west of Mauritius.
Source: (http://reefgis.reefbase.org)

[5] Measures for the protection, conservation and management of MPAs are prescribed in the amended Fisheries and Marine Resources (MPAs) Regulations of 2007.

locality. The main protagonists in this paper are the water-sports operators from the hotels and the fishers in this analysis. Other stakeholders involved relate to private pleasure craft-boat owners, residing outside Pointe-aux-Piments, and the inhabitants using the area as a recreational outlet including amateur fishing.

It was found that at the time of the interview, the age profile of the respondents pertaining to two groups of stakeholders was as follows: 53.3 per cent of the fishers were in the age group of 50–59 followed by 26.7 per cent of them falling into the category of 40–49. Among the hotel water-sports group, 63.2 per cent were found to be in the age group 20–29, followed by 29.2 per cent found within the 30–39 range. As it can be deduced, the fisher population was already ageing whereas that of the group of water-sports practitioners was relatively younger.

All the 16 fishers interviewed were male, and this tendency was also observed among all the water-sports practitioners with the exception of 1 female respondent.

4. Main findings of fieldwork

The empirical findings with respect to the different responses and strategic motives of the stakeholders relate to the theoretical model developed. Evidence on the willingness to work together and co-operate including situations of conflict on the management of the proposed park is discussed. The responses emanate from field interviews of the two stakeholder groups.

4.1. Perceptions on marine protection and conservation

The two stakeholders were actually of the opinion that the marine resources in Balaclava needed protection and that a marine park would actually be very appropriate to restore the marine environment. By and large, the majority of the respondents claimed that a marine park would be an important mechanism to protect marine resources. Furthermore, 75 per cent of the fishers and 68.4 per cent of hotel-based sports operators highlighted that it would be in their interest to protect the marine environment. Among the fishers, 81.3 per cent reported that they would encourage the development of a marine park which would also benefit the region. 90.1 per cent of the water-sports operators, claimed that they would be in favor of a marine park.

4.2. Ability to work together and co-operate

Regarding the agents' perceptions on the willingness to work with one another and with other stakeholders, it was found that hotel group found it really hard to collaborate and work with the fishermen and the converse was also true. Actually, 56.25 per cent of fishers answered positively to work in association with other stakeholders, while 47.4 per cent of the water-sports practitioners believed that this would be possible. In particular, it was found that fishers perceived it rather difficult to work with hotels but much easier to work with non-hotel-based water-sports practitioners. On the other hand, hotel-based sports operators would be more unwilling to work with the fishers than otherwise. The individual group responses are reported in Table 1 below:

Table 1
Cross-Table Indicating the Responses (in Percentage) of One Stakeholder Group on the Importance of being in Contact with Other Stakeholder Groups.

Contact important between:	Fishers	Community Members	Hotel Sports Operators	Private Water Sports Operators
Fishermen	66%	65%	34%	46%
Hotel Sports Operators	28%	55%	78%	76%

Source: Authors

In fact, this table also shows that fishers may not be very comfortable to work with their own comrades as it was found that 66 per cent would be willing to work in close collaboration but the degree of collaboration was reported lower when working with other stakeholders was considered. While 65 per cent of the fishers were willing to collaborate with the community based in Balaclava, only 34 per cent and 46 per cent among them would be willing to work with hotel sports operators and private sports operators respectively. The willingness to work with the hotels was the lowest in ratio terms. Alternatively, hotels sports operators were least willing to work with fishers. They were also more inclined and positive to work with other hotels and even private sports operators albeit less with the community. In general, the tendency shows that hotels might be more positive to work with other stakeholder groups other than with the fishers. The lack of willingness to co-operate in the eventuality that the marine park would become operational constitutes a threat to the viability of co-management and sustainability of such a park. Besides, the possibility of conflict among the two main stakeholder groups, fishers and hotels tour operators, could not be ruled out. The possible reasons provided by them on why it would be difficult to work with each other were the lack of communication which actually existed among them, the absence of a proper platform for interaction, the threat of one's return being adversely affected by any potential agreement and the uneasiness to work together.

4.3. Conflict resolution

With respect to whether there would be any form of consensus among them on any particular issue related to the marine park, should they happen to work together, the two stakeholders provided positive responses as follows; 84.2 per cent from hotel-based sports operators and 81.3 per cent from the fishers. As such, there was a general consensus among them that it would not be impossible to resolve any anticipated conflict.

4.4. Ability to manage the marine park

63.2 per cent of the hotel-based sports operators and 62.5 per cent of the fishers indicated that they had the relevant experience and expertise to help in the management of the marine park at Balaclava. Moreover, while 57.9 per cent of the sports operators revealed that they would no longer exploit marine resources as they used to do before the establishment of the marine park only 43.8 per cent of the fishers subscribed to this contention.

89.5 per cent of the water sports operators and 81.3 per cent of the fishers would be interested in co-managing the park. In addition, 94.7 per cent of the former and 81.3 per cent of the latter had shown keen interest in any capacity building that might be offered by policymakers and Non-Governmental Organisations (NGOs) to improve their management skills.

These findings simply show that it be very difficult for these two stakeholders to work together. The willingness to co-operate was found to exist but might still be very shaky. The empirical results presented in Table 1 show the lack of willingness to co-operate between the two major stakeholders. If the two groups of recreational sports operators (hotels and private) collude, it would be even more difficult for fishers to manage their activity and at their own pace. Claiming that they could both work together once the marine park would be established might be interpreted as a strategic move towards opportunistic ends. Necessarily therefore, the possibility to move to a Nash equilibrium representing unsustainability of resource exploitation is confirmed in the absence of trust and of a platform for interaction among the two and all other stakeholders.

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5. Concluding remarks and policy implications

This paper has tried to model the attitudes and perceptions of 2 major stakeholder groups, fishers and hotel water-sports operators, regarding the implementation and enforcement of a marine park in the region of Pointe aux Piments located to the North West of the island of Mauritius. A game theory framework is used for this purpose to model the different strategic moves that the players could adopt at different instances in the setting up of such a park. The model is empirically verified through survey techniques where interviews of fishers and hotel pleasure craft operators were undertaken. The findings revealed that both stakeholder groups would be willing to protect the marine environment and its ecosystem. However, the ability to actually do so would be constrained by several weaknesses found during the interviews that would jeopardize the successful management of the park. In particular, there were rather high levels of apprehension among both fishers and hotel tour operators to co-operate and work constructively. In short, this research has shown that there is no convincing evidence that potential co-management of the Balaclava Marine Park would be a success.

Some of the policy implications which emanate from this study are that, first, it would be imperative to build capacity among all the stakeholders on shared benefits and the virtues of working together. Moreover, a comprehensive action plan should be developed by the authorities, to facilitate the establishment of the marine park and to ensure sustainable livelihoods. In collaboration with NGOs, local authorities could consider the possibility of instituting capacity building programmes that would play a key role in the collective management of the park and its long term sustainability. Once legally enforced, the authorities should introduce penalties for defaulting stakeholders to prevent moral hazard for the sustainability of the park's resources.

The game theory model presented in this paper and applied to the very context of setting up of a marine park towards coastal zone management can be easily applied to other circumstances related to natural resources management involving the so-called ‘tragedy of the commons’. In particular, it might be relevant to apply this model to consider situations involving different stakeholders, with similar or competing interests, ill-defined property rights in which high economic values are involved. This would help policy makers to identify gaps and be prepared for potential areas of conflict or reluctance to co-operate.

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